

O'REILLY®

Advanced JavaScript Bootcamp

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How class works

- 3 hours today
- 10 minute breaks approximately on the hour
- Class interaction highly encouraged!



What will you get out of this course?

- How functions are first-class values
- How callbacks and higher-order functions work
- How arrow functions work
- How objects combine state and behavior
- How `this` works with functions and methods
- How to create and use JavaScript classes
- How to extend a JavaScript class
- How to use static properties and methods
- How Prototypal inheritance works

Your instructors



Elisabeth Robson



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Eric Freeman

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Prototype



The class is how we create object instances.

Player Class

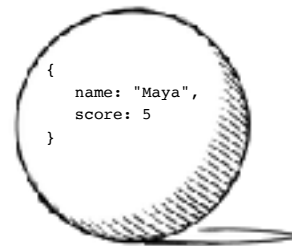
`Player`

Every class comes with a prototype object. This is where methods are stored, like `addPoint()`.

Player Prototype

`maya.addPoint()`

The object instance is where the instance properties are stored.



Player instance

`maya.name`



Static properties

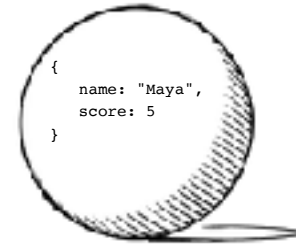
The class is how we create object instances.
This is where static properties are stored.

Player Class

Every class comes with a prototype object. This is where methods are stored, like `addPoint()`.

Player Prototype

The object instance is where the instance properties are stored.



Player instance

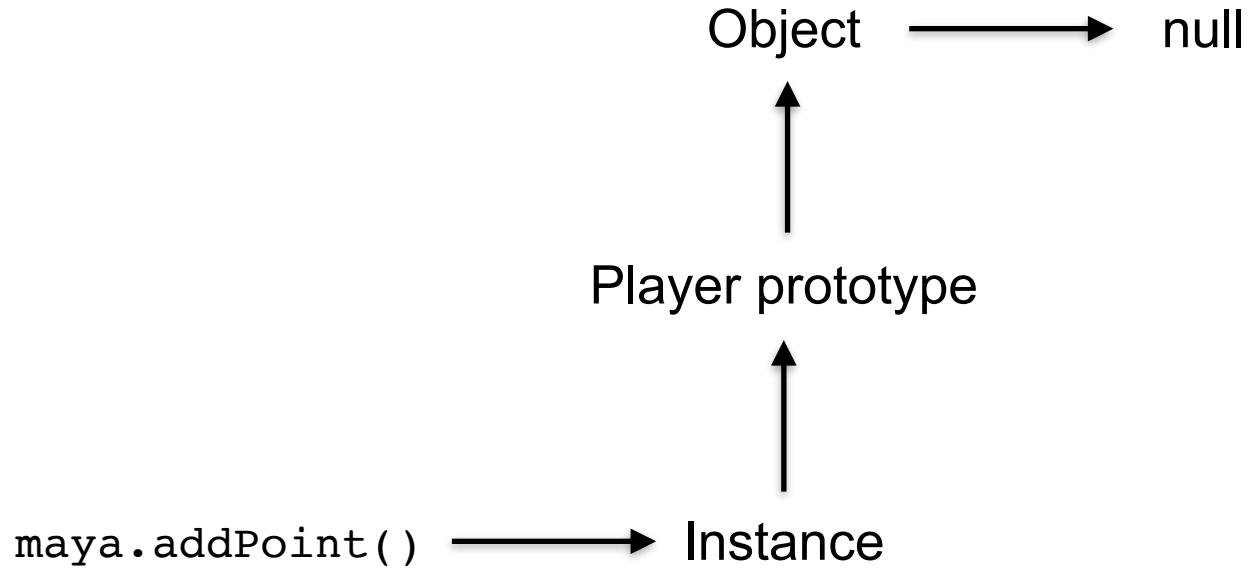
`Player.playerCount`
`Player.getPlayerCount()`

`maya.addPoint()`

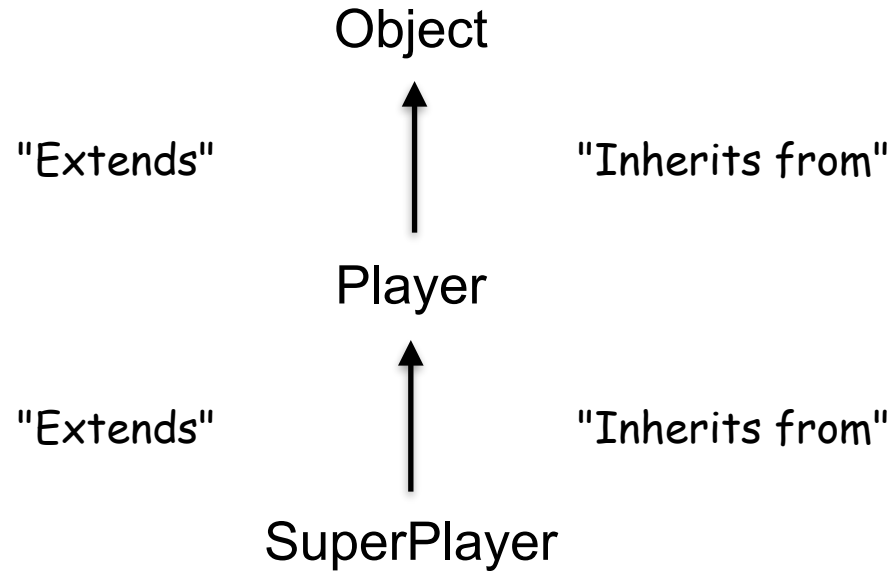
`maya.name`



Prototype look-up chain

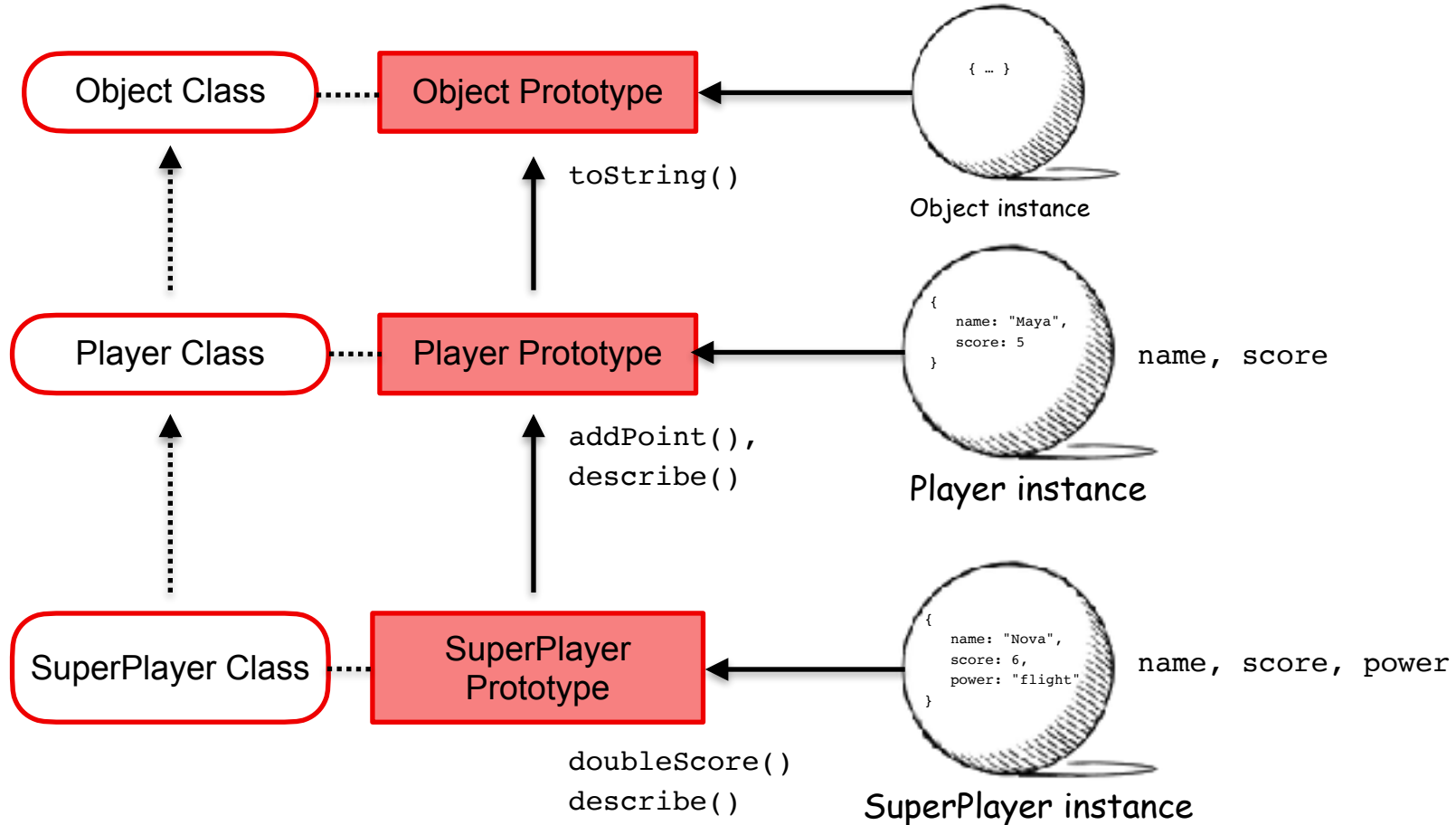


Prototypal inheritance





Prototypal inheritance look-up chain





Wrapping up

- Functions are values
- Functions can be stored, and passed behavior
- Functions include callbacks and higher-order functions
- Use arrow functions for lightweight callbacks
- Use objects with state (properties) and behavior (methods)
- We create instances from classes
- `this` selects the receiver's state
- Prototypes share and inherit behavior



Wrapping up

- There's a lot more to explore with JavaScript.
- In this course, we focused on functions and objects
 - Functions as first-class values
 - Objects made from classes with properties and methods
 - How `this` works in objects
 - How `new` and constructor functions work
 - Static properties and methods
 - Prototypal inheritance
- From here explore closures, promises, `async/await`, the event model, the concurrency model, and much more!

Thank you!

The image features the O'Reilly logo in white text on a blue background. The background has a gradient from dark blue on the left to light blue on the right. On the left side, there are several overlapping, semi-transparent blue circles of varying sizes, creating a decorative effect.

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